ThoxInux kernel verification

Released EFI v0.9.1 passed two real boots under QEMU TCG, including controller checks, embedded-domain lifecycle commands and clean shutdowns.

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Q35 / OVMF / 1 vCPU / 2 GiB / 48-bit IOMMU | QEMU 8.2.2

2	8	22	2
boots passed	controller checks	lifecycle commands	clean shutdowns

Recorded headless workflow

Verbatim harness output, lines 1-26. Display wrapping only; complete raw output is preserved in the repository. This is a document render of an actual local run.

```
cold: released ThoxInux v0.9.1 boot and four controller checks passed
cold: domain new azureprobe: passed
cold: domain set azureprobe image=embedded://minimal-system-v0 init=/sbin/init memory-mib=128 autostart=false:
passed
cold: domain start azureprobe: passed
cold: domain status azureprobe: passed
cold: ps: passed
cold: domain stop azureprobe: passed
cold: domain status azureprobe: passed
cold: domain start azureprobe: passed
cold: domain status azureprobe: passed
cold: domain restart azureprobe: passed
cold: domain status azureprobe: passed
cold: domain fail azureprobe: passed
cold: domain status azureprobe: passed
cold: domain start azureprobe: passed
cold: domain status azureprobe: passed
cold: domain stop azureprobe: passed
cold: domain status azureprobe: passed
cold: clean GINUX_KERNEL_OK shutdown passed
recovered: released ThoxInux v0.9.1 boot and four controller checks passed
recovered: domain status azureprobe: passed
recovered: domain start azureprobe: passed
recovered: domain status azureprobe: passed
recovered: domain stop azureprobe: passed
recovered: domain status azureprobe: passed
recovered: clean GINUX_KERNEL_OK shutdown passed
```

Acceptance boundary

Actual local QEMU execution of the released kernel and embedded-domain lifecycle. Azure deployment, production service, native Hyper-V, a graphical desktop and MeshStack were not tested. Rootfs pending does not establish ready Rocky Linux or systemd userspace.

Reproduce the test and inspect the proof

Use a Linux host with Python 3.11 or newer. Run from the repository root as an ordinary user after installing QEMU, OVMF and util-linux. Choose a fresh output directory.

```
sudo apt-get update
sudo apt-get install --yes qemu-system-x86 ovmf util-linux
python3 tools/verify_kernel.py --output /tmp/thoxinux-proof-01
```

1. Verify pinned release bytes. The verifier checks the downloaded EFI, release receipt and checksum manifest against reviewed pins. It also verifies the bundled upstream launcher.

2. Run the real guest twice. A fresh temporary state directory holds eight controller checks, 22 embedded-domain commands and two clean shutdowns. The second boot inspects recovered domain state.

3. Review the saved evidence. The output directory contains summary.json, lifecycle result files, supervisor receipts, complete serial.log and supervisor logs. The verifier cleans up its temporary guest processes.

4. Accept Azure separately. The deployment Quickstart creates the Azure host. Its acceptance extension must verify the live systemd service and current guest receipt before that Azure deployment is accepted.

Hashes and source identity

Released GINUX.EFI

`1bc1c7af33470edc009fa5756d3fe6063573660e07e157d19397afc680a7bb89`

Complete serial.log

`8e6fa0f40e9b02809132df99ca0363e43fc0aba83e7dbc41020220c3d5256ba9`

Harness console log

`5155517e3ad42f135c413f1b5292c7b05f5e7861d3371f4f31b0752ed7ab4d4b`

Tested portable verifier

`3825e4808a406c243ecbd8b3b2f85eefa0956edc34a84c0cb5720ef83e8fee71`

Kernel source commit

`fbad5952f20281ef6ad7eaefc1f54f7db4c0c6e9`

The kernel release is **v0.9.1**; its actual banner remains **ThoxInux 0.9.0**. Deployment package versions are separate. The final guest state was stopped. This report does not establish ready Rocky Linux/systemd userspace, a graphical desktop or Azure deployment.

Recorded sources

summary.json - Machine-readable result and exact scope
harness-console.log - Full harness transcript
serial.log - Complete serial output
readback-validation.json - Log-slice hash checks and process cleanup
github.com/Thox-ai/thoxinux-azure | kernel release: Thox-ai/thoxinux v0.9.1

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